```
default 172.16.4.1 0.0.0.0     UG    0    0    0 eno 16777984
172.16.4.0  0.0.0.0 255.255.252.0  U 0    0    0 eno 16777984
172.16.200.0 0.0.0.0 255.255.252.0 U 0    0    0 eno 16777984
```

The above output of *netstat –r* with IP addresses indicates that the name server of the target network is not functioning properly or has some configuration issues.

After finding out the basic configuration, one can explore all the connected hosts with respect to each subnet identified by the router table entries. There are two useful commands, *ip* and *arp*, with the help of which it is possible to get data on all the close neighbours, along with their respective MAC addresses. The administrative command *ip* is so powerful that it is also possible to use it for manipulating routing devices, and editing routing and tunnelling policies. While *arp* is good to identify the connected hosts with their MAC addresses and connection types, *ip* can show you the status of the connections.

```
>arp
Address      HWtype  HWaddress        Flags Mask      Iface
172.16.6.83  ether   34:17:eb:b4:9b:6d  C        eno16777984
172.16.4.66  ether   e4:11:5b:4b:d0:63  C        eno16777984
172.16.5.96  ether  a0:48:1c:9e:72:02   C        eno16777984
172.16.5.230 ether  e0:cb:4e:ff:7e:bd   C        eno16777984
172.16.5.213 ether a0:b3:cc:e8:e4:16    C        eno16777984
172.16.7.226 ether 6c:62:6d:d2:1e:70    C        eno16777984
172.16.5.60  ether  10:78:d2:53:1b:ee   C        eno16777984
172.16.7.142 ether 08:ec:a9:ec:b1:9d    C     eno16777984
172.16.5.241 ether 04:7d:7b:fa:b7:c0    C     eno16777984
```

The network management command *ip* with the option *neigh* provides more detailed information than the other network management and exploration commands.

```
>ip neigh
172.16.6.83 dev eno16777984 lladdr 34:17:eb:b4:9b:6d STALE
172.16.4.66 dev eno16777984 lladdr e4:11:5b:4b:d0:63 STALE
172.16.5.96 dev eno16777984 lladdr a0:48:1c:9e:72:02 STALE
172.16.5.230 dev eno16777984 lladdr e0:cb:4e:ff:7e:bd
REACHABLE

172.16.5.213 dev eno16777984 lladdr a0:b3:cc:e8:e4:16 STALE
172.16.7.226 dev eno16777984 lladdr 6c:62:6d:d2:1e:70 STALE
172.16.5.60 dev eno16777984 lladdr 10:78:d2:53:1b:ee STALE

172.16.7.142 dev eno16777984 lladdr 08:ec:a9:ec:b1:9d STALE
172.16.5.241 dev eno16777984 lladdr 04:7d:7b:fa:b7:c0 STALE

172.16.200.88 dev eno16777984 lladdr 00:16:c7:2c:3f:bf STALE
172.16.4.54 dev eno16777984 lladdr 00:15:5d:07:55:0b
REACHABLE

172.16.6.108 dev eno16777984 lladdr 34:17:eb:b4:9c:01 STALE
172.16.200.90 dev eno16777984 lladdr 00:16:c7:2c:3f:bf STALE
172.16.5.243 dev eno16777984 lladdr 20:47:47:1f:5e:2f STALE
```

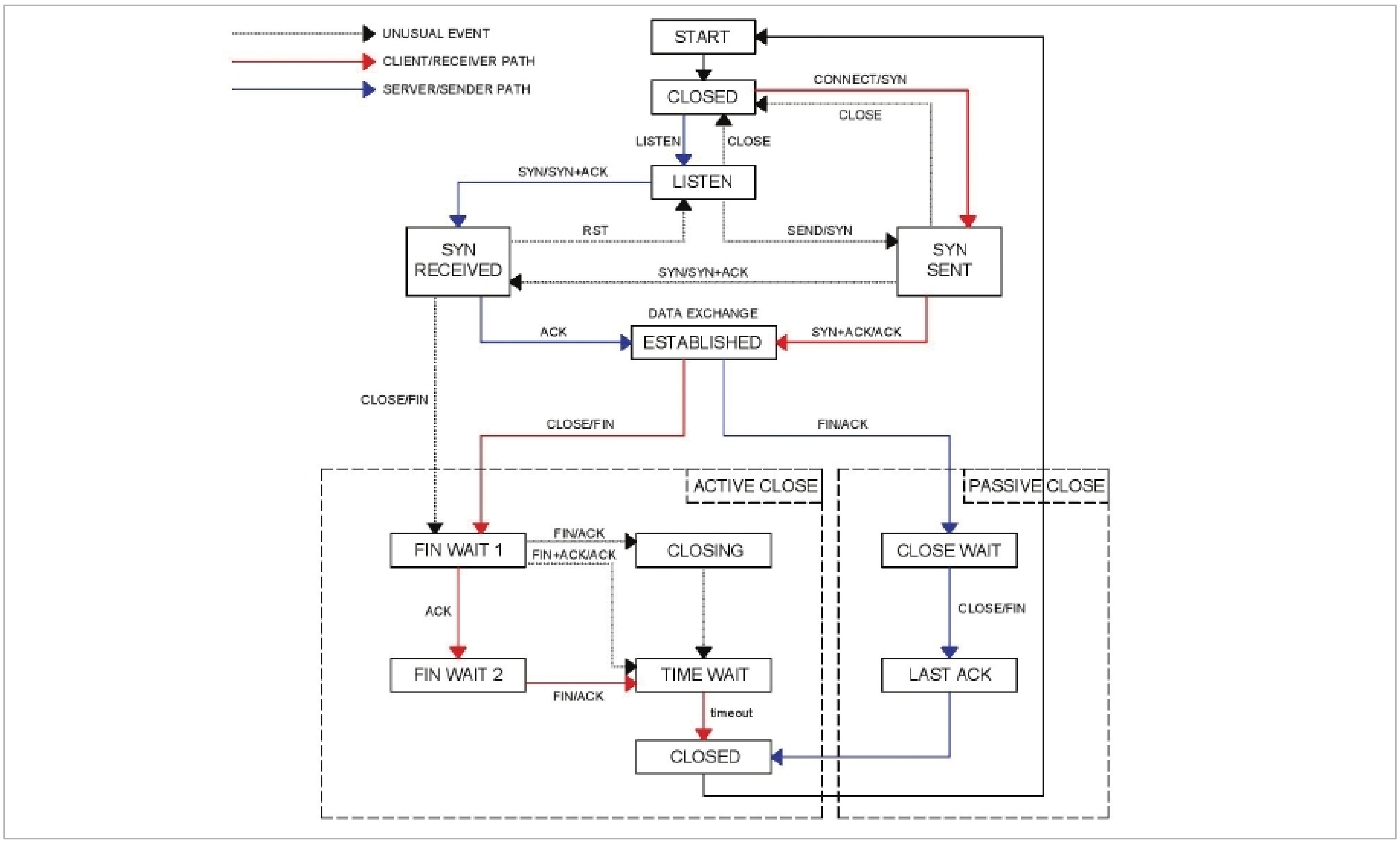

Figure 2: Socket client server connectivity TCP state diagram (Ref 1)